

On a geometric interpretation of the $1/4$ factor in black hole entropy

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Abstract. The Bekenstein–Hawking entropy $S = k_B A / 4\ell_P^2$ contains a coefficient $1/4$ that has been calculated by many approaches to quantum gravity. We show that this factor arises from the causal geometry of any null boundary in a four-dimensional Lorentzian spacetime. A heuristic accessibility argument is supported by a purely geometric derivation using the canonical symplectic structure on the cotangent bundle and the causal splitting of the space of null rays. No gravitational field equations are invoked.

1. Introduction

The entropy of a black hole is

$$S = \frac{k_B A}{4 \ell_P^2}, \quad \ell_P^2 = \frac{G \hbar}{c^3}, \quad (1)$$

where for the remainder of this paper $k_B = 1$, and the other physical constants—($\hbar, G, c$)—are kept explicitly but appear only through the Planck-scale ℓ_P . The universality of the coefficient 1/4 suggests a geometric origin independent of microscopic details, as it arises in diverse frameworks including semiclassical gravity, Noether-charge methods, Euclidean path integrals, thermodynamic arguments, and combinatorial approaches [1, 2, 3, 4, 5, 6, 7].

We argue that this factor is the fraction of phase space accessible across any null causal boundary in a four-dimensional Lorentzian spacetime. We do not attempt a microscopic derivation of black hole entropy; instead, we identify a kinematic mechanism in null causal geometry that produces the factor 1/4 independently of any particular dynamics or UV completion.

2. Heuristic accessibility argument

We begin with a heuristic overview of the reduction; the formal derivation follows in Section 3.

A point in the phase space of a four-dimensional Lorentzian spacetime is specified by four coordinates and four conjugate momenta—eight dimensions in total.

(1) Null mass-shell constraint[‡] ($8 \rightarrow 7 \rightarrow 6$). Imposing a mass-shell or null condition $Q = g^{ab} p_a p_b = 0$ makes one momentum a function of the others, and the resulting evolution sweeps out orbits along the constraint surface, removing the conjugate coordinate as an independent degree of freedom. Two dimensions are lost; six remain (in the GR formalism, this is given in [8]).

(2) Null surface ($6 \rightarrow 4$). Confinement to a null hypersurface $\mathcal{N}$ in the base manifold fixes the transverse coordinate, removing one degree of freedom. In Lorentzian signature, the normal to a null surface is also tangent to it—a feature with no Riemannian analogue—so the generator direction within $\mathcal{N}$ is not independent but already determined, removing a second degree of freedom. What remains is a two-dimensional spacelike cross-section $\mathcal{S}$, carrying two canonical pairs (recovered in GR in [8, 9]).

(3) Conscripted time ($4 \rightarrow 2$). The two canonical pairs on $\mathcal{S}$ include directions tangent to the generators of $\mathcal{N}$. Deformations along these directions slide points along the same null rays without changing the surface; they carry no independent

[‡] In reparametrization-invariant language, $Q = 0$ acts as a Hamiltonian constraint generating the geodesic flow.

information. One coordinate is therefore conscripted to track position along the generators: the induced metric on $\mathcal{S}$ is positive definite, so no geometric evolution parameter exists, and one pair must serve as a relational clock for the other [10, 11, 12]. In the language of the method of characteristics, the time coordinate drops out— $F(q_1, p_1, q_2, p_2, t) \rightarrow F(q_1, p_1, q_2, p_2) = C$ —and q_2 inherits the role of classical time, parametrizing the family of solutions while p_2 selects among them. One pair is consumed; one survives—the equivalent of “dressing time” in [13].

(4) Causal restriction ($2 \rightarrow 1$ observable). A null hypersurface is a one-way causal boundary. After the clock conscription, only one canonical pair survives. This pair carries both future-directed and past-directed data, but an observer confined to one side receives signals from only one of these two components. Only half the pair is accessible (recovered in GR in [14]).

The accessible fraction of the naive surface phase space is thus

$$\frac{\frac{1}{2} \text{ pair}}{2 \text{ pairs}} = \frac{1}{4}. \quad (2)$$

Illustration. Schwarzschild coordinates are singular at the horizon and therefore unsuitable for a rigorous treatment (Section 5 uses regular Eddington–Finkelstein coordinates), but they make the dimensional reduction vivid. In these coordinates, $ds^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - r_s/r$. The null condition $Q = 0$ constrains the momenta. The null surface sits at $r = r_s$, where $f = 0$. There $ds^2 = 0$ implies that $t \rightarrow t + \epsilon$ for arbitrary ϵ leaves the geometry unchanged: the time coordinate loses all local meaning. Steps (1) and (2) have removed (t, p_t) and (r, p_r) ; what remains is the Riemannian 2-sphere $(\theta, \varphi, p_\theta, p_\varphi)$ —two canonical pairs. Step (3) conscripts one angular coordinate as the evolution parameter, leaving a single pair. Step (4) halves it.

This argument is heuristic; a purely geometric formal realization is given next.

3. Formal geometric derivation

Throughout this section (M, g) is a four-dimensional time-oriented Lorentzian manifold[§], so that $\dim T^*M = 8$.

The reduction begins from the full cotangent bundle T^*M rather than from the null cone directly for self-containment and clarity. Each step is then geometrically explicit, and the origin of every factor of 2 is transparent.

The canonical one-form on T^*M and the null constraint $Q = g^{ab} p_a p_b$ involve different types of contraction: the former uses only the natural dual pairing between vectors and covectors, while the latter requires the spacetime metric. This distinction is fundamental: the symplectic structure is a property of the cotangent bundle alone, requiring no metric, while the null condition is a statement about the spacetime

[§] For any n -dimensional manifold M , the cotangent bundle T^*M has dimension $2n$. In our case $\dim M = 4$, so $\dim T^*M = 8$.

geometry. These two structures are logically independent [15, 16]. Throughout, indices are raised and lowered using g_{ab} only when explicitly indicated.

We now formalize the heuristic argument using the symplectic geometry of null covectors.

3.1. Canonical phase space

On T^*M define the canonical one-form and symplectic form^{||}

$$\theta = p_a dx^a, \quad \omega = -d\theta = dx^a \wedge dp_a. \quad (3)$$

Define the quadratic function on T^*M

$$Q(x, p) = g^{ab}(x) p_a p_b. \quad (4)$$

The condition $Q = 0$ is not a dynamical mass-shell constraint but a geometric condition selecting the null covectors in T^*M , determined entirely by the Lorentzian metric.

The null covectors form the hypersurface

$$\mathcal{C} = \{Q = 0\} \subset T^*M. \quad (5)$$

The symplectic structure ω associates to Q a vector field X_Q —the Hamiltonian vector field of Q —on T^*M via $\iota_{X_Q}\omega = dQ$. On $\mathcal{C}$ the integral curves of X_Q satisfy

$$\frac{dx^a}{d\lambda} = \frac{\partial Q}{\partial p_a} = 2g^{ab}p_b, \quad (6)$$

where λ is the curve parameter (not a time coordinate). The tangent vector is proportional to the covector p_a with index raised by the metric. On $\mathcal{C}$ this vector is null; the integral curves are null geodesics.

3.2. Space of null rays

Since Q is constant on $\mathcal{C}$, any vector V tangent to $\mathcal{C}$ satisfies $dQ(V) = 0$. But $\iota_{X_Q}\omega = dQ$ by definition, so

$$\omega(X_Q, V) = 0 \quad \text{for all } V \in T\mathcal{C}. \quad (7)$$

That is, X_Q lies in the kernel of $\omega|_{\mathcal{C}}$.

Moreover, X_Q is tangent to $\mathcal{C}$: the flow preserves Q because $X_Q(Q) = \omega(X_Q, X_Q) = 0$ by antisymmetry.

Dimension counting:

$$\dim T^*M = 8, \quad (8)$$

$$Q = 0 \Rightarrow \dim \mathcal{C} = 7, \quad (9)$$

$$\text{quotient by } X_Q \Rightarrow 6. \quad (10)$$

^{||} We follow the sign convention of [16], with $\omega = -d\theta$.

Quotienting $\mathcal{C}$ by the one-dimensional orbits of X_Q yields the space $\mathcal{R}$ of unparametrized null geodesics (null rays). The restriction $\omega|_{\mathcal{C}}$ has a one-dimensional kernel spanned by X_Q ; it therefore descends to a non-degenerate closed 2-form on the 6-dimensional quotient $\mathcal{R}$, making $\mathcal{R}$ a symplectic manifold [8].

3.3. Restriction to a cross-section

Let $\mathcal{N}$ be a null hypersurface and $\mathcal{S} \subset \mathcal{N}$ a closed spacelike 2-surface that serves as a cross-section of $\mathcal{N}$. Each point of $\mathcal{S}$ determines a unique generator of $\mathcal{N}$, which is a single point in $\mathcal{R}$. The generators through $\mathcal{S}$ therefore define a 2-dimensional submanifold $G \subset \mathcal{R}$.

The restriction $\omega|_G$ vanishes: deformations within the family of generators of a null hypersurface carry no symplectic flux [8, 9, 17]. Hence G is isotropic[¶].

The symplectic complement $TG^\perp$ —the set of all directions in $T\mathcal{R}$ that have vanishing symplectic product with every vector tangent to G —has dimension $6 - 2 = 4$, i.e. two canonical pairs. These encode all phase-space data in $\mathcal{R}$ coupled to the surface and constitute the naive phase-space content of a surface element.

Because G is isotropic, $TG \subset TG^\perp$. Quotienting by TG removes the two directions tangent to G , leaving

$$\dim(TG^\perp/TG) = 4 - 2 = 2, \quad (11)$$

i.e. one canonical pair per surface element. This pair corresponds to the expansion–shear conjugate structure identified in [9, 18]. The quotient $TG^\perp/TG$ is the kinematic analogue of the *dressing time* construction of Ciambelli, Freidel, and Leigh [13]. The reduction is mandatory: the induced metric on $\mathcal{S}$ is positive definite, so Hamilton’s equations on $T^*\mathcal{S}$ admit no geometric evolution parameter. In Rovelli’s language [10], the two canonical pairs are partial observables; only their correlations are complete observables, so one pair must serve as a relational clock for the other. Entropy counts independent phase-space labels; a pair serving as a relational clock is not an independent label of the state but a parametrization of correlations, and therefore does not contribute to the microstate count. This mechanism is independent of the Raychaudhuri equation and of any gravitational dynamics [19].

3.4. Causal halving

Time orientation splits $\mathcal{C}$ into future-pointing ($\mathcal{C}^+$) and past-pointing ($\mathcal{C}^-$) null covectors. Since $Q(x, -p) = Q(x, p)$, the map $(x, p) \mapsto (x, -p)$ preserves $\mathcal{C}$ and maps orbits of X_Q to orbits, so it descends to an involution on $\mathcal{R}$ that exchanges the future-pointing and past-pointing components $\mathcal{R}^\pm$.

[¶] A submanifold G is called *isotropic* by analogy with the metric case: just as a null vector v satisfies $g(v, v) = 0$, the tangent vectors of G satisfy $\omega(u, v) = 0$ for all $u, v \in TG$ —they are isotropic with respect to the symplectic form. This is a symplectic condition, distinct from but motivated by the metric notion of isotropy.

Under $p_a \mapsto -p_a$ the symplectic form transforms as $\omega \mapsto -\omega$, so the reduced symplectic form on $\mathcal{R}$ also reverses sign. The Liouville measure [16], being proportional to $|\omega_{\mathcal{R}}^3|$, is invariant:

$$\mu(\mathcal{R}^+) = \mu(\mathcal{R}^-) = \frac{1}{2} \mu(\mathcal{R}). \quad (12)$$

This splitting is dynamically empty. The flow of X_Q is time-reversal symmetric not by virtue of being a Hamiltonian flow in general, but because Q is a geometric constraint on the cotangent bundle whose quadratic structure in p gives $Q(x, -p) = g^{ab}(-p_a)(-p_b) = Q(x, p)$. The map $(x, p) \mapsto (x, -p)$ therefore preserves $\mathcal{C}$ and maps orbits of X_Q to orbits. The full phase space is causally accessible to a freely falling observer who crosses $\mathcal{N}$. A null hypersurface is the only geometric structure in Lorentzian spacetime that locally partitions a neighborhood into causally disconnected regions across which signals propagate in only one direction. As such, it defines a causal boundary. It divides a neighborhood of N into two regions: future-directed null rays from S propagate into one region, while past-directed rays propagate into the other. An observer confined to one side receives signals from only one type of rays.

The causal structure of (M, g) —encoded in the distinction between the causal future $J^+(\mathcal{N})$ and causal past $J^-(\mathcal{N})$ —therefore restricts the *observable* portion of the phase space to one temporal component. At each surface element, the single canonical pair identified in Section 3.3 carries both $\mathcal{R}^+$ and $\mathcal{R}^-$ data, but only one component is accessible:

$$\mu_{\text{obs}} = \frac{1}{2} \mu_{\text{full}}. \quad (13)$$

The full phase space remains intact. The halving is not a dynamical reduction but a causal restriction on observational access, determined entirely by the Lorentzian causal structure of $\mathcal{N}$.

4. Entropy

We now collect the reduction chain. Starting from eight phase-space dimensions per spacetime point on T^*M : the null constraint $Q = 0$ removes one (Section 3.1); the quotient by X_Q removes another (Section 3.2); the symplectic complement $TG^\perp$ selects the four dimensions coupled to the surface—two canonical pairs—which constitute the naive phase-space content of a surface element (Section 3.3); the isotropic quotient $TG^\perp/TG$ removes the remaining two, leaving one canonical pair.

Time orientation (Section 3.4) divides the surviving pair into future and past components of equal measure. The full phase space remains intact, but a causal boundary permits observation of only one temporal direction. An observer on one side therefore has access to half of the surviving pair.

According to common wisdom, entropy counts microstates consistent with the observed macrostate. For an exterior observer, the macrostate is defined by causally accessible data alone; microstates in $\mathcal{R}^-$ are causally inaccessible and therefore logically

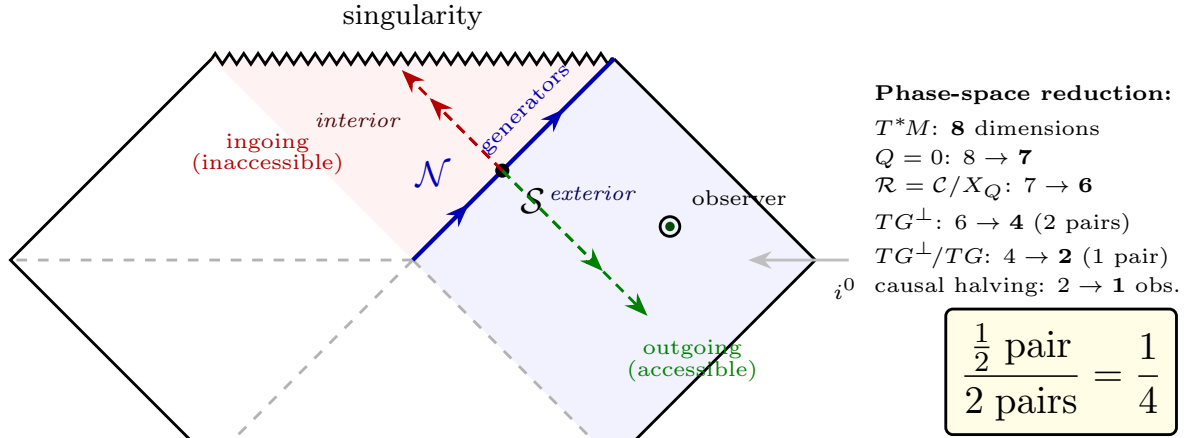


Figure 1. Penrose diagram of a Schwarzschild black hole with the phase-space reduction chain. The cross-section $\mathcal{S}$ on the future horizon $\mathcal{N}$ is traversed by null rays in two temporal directions. Outgoing rays (green, dashed) reach the exterior observer; ingoing rays (red, dashed) fall into the interior and are causally inaccessible. The reduction chain (right) traces the eight cotangent-bundle dimensions to a single observable degree of freedom, yielding the factor $1/4$.

excluded from the count. No coarse-graining or tracing-out is required: the exclusion follows directly from the definition of the macrostate and halves the accessible count.

The observable fraction of the naive surface phase space is

$$\frac{\frac{1}{2} \text{ pair}}{2 \text{ pairs}} = \frac{1}{4}. \quad (14)$$

Whatever Planck-area cell count $N = A/\ell_{\text{P}}^2$ the UV completion provides—whether from combinatorial tiling [7] or any other microscopic framework—the combinatorics supplies the count while the geometry fixes the accessible fraction, yielding

$$S = \frac{A}{4 \ell_{\text{P}}^2}. \quad (15)$$

5. Application to black holes

To make contact with the black hole literature, we now map each step of the general construction onto a Schwarzschild horizon in ingoing Eddington–Finkelstein coordinates

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dv^2 + 2 dv dr + r^2 d\Omega^2, \quad (16)$$

where v is the advanced time and the metric is regular at $r = r_s$. The argument applies to any null boundary, but the Schwarzschild case provides a concrete identification of every reduction step (Fig. 1).

Phase space. The cotangent bundle T^*M has eight dimensions: $(v, r, \theta, \varphi, p_v, p_r, p_\theta, p_\varphi)$.

Null constraint ($8 \rightarrow 7$). Imposing $Q = g^{ab}p_ap_b = 0$ restricts the momenta to the null cone. This selects the hypersurface $\mathcal{C} \subset T^*M$.

Quotient by X_Q ($7 \rightarrow 6$). The flow of X_Q moves a point along a null geodesic to a different affine parameter value. Quotienting identifies all parametrizations of the same geodesic, yielding the space of null rays $\mathcal{R}$.

Horizon as null hypersurface. The event horizon $\mathcal{N}$ is the surface $r = r_s$. In these coordinates the metric is non-degenerate there and the normal covector $n_a = (dr)_a$ is null: $g^{ab}n_an_b = g^{rr} = 0$ at $r = r_s$. Because n^a is both normal and tangent to $\mathcal{N}$, it defines a preferred direction within the horizon: the generator direction, which coincides with $\partial/\partial v$ at $r = r_s$.

Foliation and cross-section ($6 \rightarrow 4$). The horizon is foliated by generators—curves at constant (θ, φ) running in the v -direction. Quotienting $\mathcal{N}$ by its generators yields a 2-sphere $\mathcal{S}$ with coordinates (θ, φ) . In $\mathcal{R}$, this corresponds to the isotropic submanifold G , and its symplectic complement $TG^\perp$ has dimension $6 - 2 = 4$: two canonical pairs. These are the naive phase-space data of a surface element of $\mathcal{S}$.

Dressing time ($4 \rightarrow 2$). The isotropic quotient $TG^\perp/TG$ removes the generator-labeling directions, consuming one canonical pair. The surviving pair corresponds to the expansion Θ and shear σ_{ab} of the null congruence—the Bondi news data on the horizon [9].

Causal halving ($2 \rightarrow 1$ **observable**). An observer at $r > r_s$ receives signals carried by outgoing null rays only. The ingoing component crosses the horizon and is causally inaccessible from the exterior. The full phase space is intact—an infalling observer encounters nothing special at $r = r_s$ —but the exterior observer has access to only half of the surviving canonical pair.

The observable fraction is $\frac{1}{2}$ of one pair out of a naive two: $1/4$. Each step in the general construction of Sections 3–4 has a direct black hole counterpart, but none of the steps depends on the Einstein equations. The coefficient $1/4$ emerges from the null causal structure of the horizon, not from the dynamics that produces it.

6. Discussion

The coefficient $1/4$ is a consequence of null causal structure and time orientation in four-dimensional Lorentzian spacetime. It is independent of the Einstein equations, of any gravitational action, of Newton’s constant (except through the Planck length in defining the cell size), and of any particular spacetime solution.

The derivation is explicitly four-dimensional; the generalization to arbitrary D is under investigation. The argument requires three ingredients: a Lorentzian metric, null geodesics, and a closed 2-surface that acts as a one-way causal boundary. These are shared by every framework in which black hole entropy has been computed. The universality of the coefficient across semiclassical, string-theoretic, and loop gravity

derivations is therefore not a coincidence to be explained by each framework separately, but a geometric fact that precedes all of them.

Epistemic interpretation. The entropy in this derivation counts accessible phase-space volume, not missing information. No degrees of freedom are traced over, integrated out, or lost. The full phase space of the null surface is intact; the causal boundary restricts which portion an exterior observer can probe. The factor $1/4$ is the geometric measure of this epistemic restriction for a null causal boundary.

What this derivation does and does not explain. The Planck-area cell count $N = A/\ell_{\text{P}}^2$ is an input, not a result, and null geodesics are not identified here as the microscopic degrees of freedom responsible for black hole entropy. This paper derives the fraction $1/4$ but not the proportionality to area, which requires a UV completion that discretizes the surface into Planck-scale cells (see, e.g., [7]). The reduction chain does, however, offer a geometric explanation for why entropy is naturally associated with a two-dimensional surface: the successive constraints reduce all observable phase-space degrees of freedom to those living on the spacelike cross-section $\mathcal{S}$, providing a kinematic basis for the holographic association of entropy with area. The argument is kinematic: whatever microscopic framework supplies the state count—field modes, spin network states, or combinatorial surface elements—the null causal geometry fixes the accessible fraction to $1/4$. The derivation does not count those degrees of freedom, explain why black holes radiate, why the radiation is thermal, or how information is returned. In this sense, the derivation is theory-agnostic: the factor $1/4$ is a consequence of null causal geometry alone, independent of whether the underlying microscopic theory is quantum field theory in curved spacetime, string theory, loop quantum gravity, or any other UV completion.

Relation to the information paradox. The causal halving step implies that the phase-space data behind the horizon is inaccessible, not absent. An infalling observer encounters the full phase space; the restriction is perspectival, consistent with complementarity and with the expectation that unitarity is preserved globally.

Conflict of Interest

The author declares no competing interests.

Data Access Statement

This work is purely analytical. No experimental data were generated or analysed.

Ethics Statement

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